Software Requirements Specification

Haat_Bazar Architecture & Functional System Definition

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Date: October 13, 2026

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) outlines the architectural, functional, and operational requirements for **Haat_Bazar**, a digital multi-sided agricultural marketplace. It covers the complete lifecycle of farmer-to-consumer and farmer-to-wholesaler transactions.

1.2 Document Conventions

- **Priority Ratings:** Requirements are marked as High, Medium, or Low priority to guide iterative development sprints.
- **Formatting:** Standard markdown syntax is utilized throughout, complimented by structured visual diagrams and explicit data schema tables.

1.3 Intended Audience and Reading Suggestions

This document is intended for software engineers, project managers, UI/UX designers, and supply chain stakeholders. It is recommended to read Section 2 (*Overall Description*) to understand the core marketplace dynamics before proceeding to the detailed technical specifications in Section 4 (System Features).

1.4 Product Scope

Haat_Bazar bridges the operational gap between rural farmers and prospective buyers (both B2C consumers and B2B wholesalers) by acting as a transparent digital middleman. The platform aims to eliminate exploitative intermediary layers in the traditional agricultural supply chain. By enforcing Minimum Order Quantities (MOQs), dynamic routing, and flat service charges, it ensures farmers avoid operating losses while buyers receive verified, fresh produce.

1.5 References

- Haat_Bazar Operational Mathematical & Pricing Models (Oct 2026)
- Haat_Bazar System Architecture & Microservices Specifications (Oct 2026)

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2. Overall Description

2.1 Product Perspective

Haat_Bazar operates as a standalone digital ecosystem composed of multi-tenant user-facing applications (native mobile and web interfaces) backed by internal hub-management and logistics-routing microservices. It centralizes historically fragmented agricultural trade into a unified, traceable digital workflow.

2.2 Product Functions

- Multi-Role Registration & Verification (Farmers, Consumers, Wholesalers)
- Harvest Batch Inventory Listing & Lifecycle Tracking
- Dynamic MOQ and Automated Cart Validation Engines
- Integrated Escrow Payment Processing & Automated T+1 Settlements
- Local Hub Physical Inspection & Quality Grading (Class A, B, and C)
- Multi-Stop Direct Logistics and Direct Delivery Routing

2.3 User Classes and Characteristics

User Role	Target Audience	Key Characteristics & Requirements
Farmers (Sellers)	Rural agricultural producers	Low digital literacy. Requires localized UI, voice guidance, icon-heavy navigation, and simplified crop listing inputs.
Consumers (B2C)	Urban household buyers	High digital literacy. Expects seamless e-commerce UX, rapid checkout, clear MOQ thresholds, and transparent fee breakdowns.
Wholesalers (B2B)	Commercial institutions	High-volume purchasing logic. Requires bulk pricing tiers, scalable delivery options, dynamic invoices, and explicit MOQs.
Hub Operators	Regional logistics personnel	Operations-focused. Needs rapid batch scanning, physical quality grading input tools, and manifest verification workflows.

2.4 Operating Environment

- **Backend Infrastructure:** Python/Django, PostgreSQL primary datastore, Redis caching layer, Docker container orchestration.
- **Client Applications:** React Native (Cross-platform Mobile), TypeScript/React (Web Administration Portal).
- **Target Operating Systems:** iOS 14+, Android 8.0+, modern evergreen web browsers.

2.5 Design and Implementation Constraints

- **Network Connectivity:** Must support offline-first state synchronization and localized SMS transactional confirmations for rural regions with unreliable cellular coverage.
- **Localization:** Application UI, system prompts, audio instructions, and invoice structures must fully support the Bengali language natively.

2.6 User Documentation

Interactive visual walkthroughs and audio-assisted tutorials are embedded within the Farmer application. Standard tooltip overlays, comprehensive FAQ modules, and automated support flows are integrated into the buyer ecosystem.

2.7 Assumptions and Dependencies

- Integration availability with local Mobile Financial Services (MFS) including bKash and Nagad for transaction clearances.
- Regional road connectivity and local transport networks permit same-day or next-day delivery from regional collection hubs to urban delivery points.

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3. External Interface Requirements

3.1 User Interfaces

- **Farmer Application:** High-contrast, icon-driven interface utilizing large tap targets and minimal text input fields to ensure accessibility across varying literacy levels.
- **Buyer Application:** Standard e-commerce layout featuring real-time visual progress indicators for cart thresholds (e.g., "Add 2.5kg of produce to meet the minimum delivery threshold").

3.2 Hardware Interfaces

- Handheld optical QR/Barcode scanners deployed at physical collection hubs for rapid inventory intake and tracking.
- Mobile device cameras integrated for visual crop verification and image analysis workflows.

3.3 Software Interfaces

- Mobile Financial Services (MFS) RESTful API gateways for wallet-based payments and disbursements.
- Google Maps Platform APIs for geographic distance calculation, route optimization, and delivery fee calculations.

3.4 Communications Interfaces

- HTTPS with TLS 1.3 encryption for secure API client-server communication.
- Automated SMS Gateway integrations for operational alerts, payment confirmations, and dispatch notifications.

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4. System Features

4.1 Order Constraint Engine (MOQ Guard)

4.1.1 Description and Priority

Priority: High. Protects supply chain economics by ensuring every generated order fulfills minimum weight criteria required for cost-effective hub handling and transport.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** Buyer adds 1kg of tomatoes to the active shopping cart.
- **Response:** System computes total cart weight, identifies deficit, and restricts checkout progression with an active prompt: *"Minimum regional basket weight is 5kg. Please add 4kg of additional produce to proceed."*
- **Stimulus:** Buyer adds 4kg of potatoes.
- **Response:** System validates weight threshold, unlocks cart, and enables checkout transition.

4.1.3 Functional Requirements

Requirement ID	Specification Description	Priority
REQ-1	The system shall continuously evaluate shopping cart item weights against dynamic thresholds and restrict checkout until criteria are fully met.	High
REQ-2	The system shall enforce distinct, configurable minimum order thresholds (e.g., 50kg) for	High

Requirement ID	Specification Description	Priority
	verified Wholesaler accounts to unlock wholesale pricing.	

4.2 Dynamic Fee & Commission Engine

4.2.1 Description and Priority

Priority: High. Provides automated, fully transparent fee calculation without unverified third-party markups.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Buyer initiates checkout sequence.
- **Response:** System fetches the base crop price from the farmer listing, queries routing distances via geolocation APIs, calculates service fees, and renders the itemized price breakdown.

4.2.3 Functional Requirements

Requirement ID	Specification Description	Priority
REQ-3	The system shall automatically compute total transaction cost using the formula: `Final Total = Base Crop Price + Platform Service Fee + Logistics/Delivery Fee`.	High
REQ-4	System administrators shall have permissions to configure platform service percentage fees (3% to 5%) granularly per crop category.	Medium

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Marketplace inventory search and category page loads must complete in under 500ms under standard network conditions.
- System backend must scale dynamically to handle 10x traffic bursts during peak morning harvest listing hours.

5.2 Safety Requirements

- Full batch-level traceability must be maintained, linking every fulfilled order back to the specific farmer and harvest batch for food safety auditability.

5.3 Security Requirements

- Implementation of strict Role-Based Access Control (RBAC), ensuring operational Hub personnel cannot edit farmer payout account details or override base pricing.
- End-to-end AES-256 encryption for stored user Personally Identifiable Information (PII) and location data.

5.4 Software Quality Attributes

- **Usability:** System design must accommodate low-literacy users through visual cue icons and optional voice navigation.
- **Reliability:** Minimum core system operational availability of 99.9% for listing, payment processing, and checkout modules.

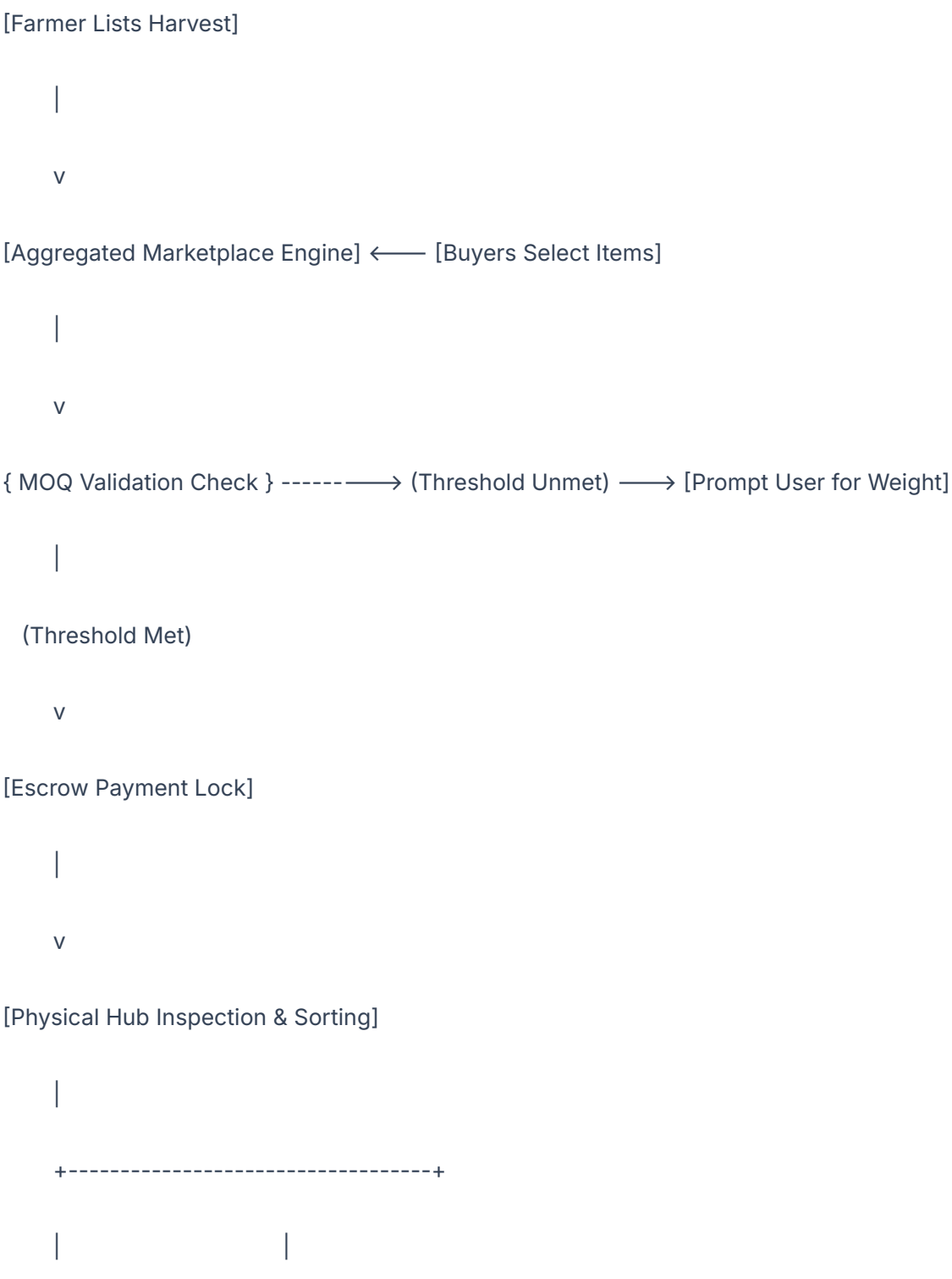
5.5 Business Rules

- **T+1 Settlement Policy:** Payouts to farmer MFS accounts must automatically process within 24 hours (T+1) following physical inventory verification and grading by a certified Hub Operator.

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6. Visualizations & Analysis Models

6.1 System Transaction Data Flow



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[Dispatched for Last-Mile Delivery] [Automated T+1 Payout to Farmer]

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[Order Delivered & Completed]

6.2 Transparent Pricing Model Breakdown

Cost Component	Recipient / Beneficiary	Calculation & Allocation Logic
Base Crop Price	Farmer	100% of the unit price set directly by the producing farmer.
Platform Service Fee	Haat_Bazar Platform	Fixed percentage fee (3-5%) applied to cover digital operations, hub logistics overhead, and escrow management.
Logistics & Delivery Fee	Shipping Partners	Variable cost calculated based on origin hub distance, destination, and order weight.
Total Order Value	Buyer Payment	Aggregated single payment sum collected transparently at point of checkout.